

Integrating Census Data with Electronic Health Records to Assess the Impact of Social Determinants of Health on Opioid Use Disorder

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Abstract

Electronic health record (EHR) data has become an invaluable resource for biomedical research, yet it is limited for health disparity and equity research due to the lack of comprehensive information on social determinants of health (SDoH). In this study, we enriched EHR data from a large healthcare system in the St. Louis metropolitan area with SDoH from United States Census data to examine patterns of opioid use disorder (OUD) and access to medications for OUD (MOUD). By linking patient addresses in the EHR to census tracts in the geospatial data, we identified geographic and sociodemographic disparities in OUD prevalence and MOUD treatment allocation. Patients who were male, Black, or living in areas with greater poverty remained less likely to receive treatment. These findings highlight the value of integrating EHR and geospatial SDoH data to uncover inequities in care and inform more equitable, targeted strategies to address the opioid epidemic.

Introduction

The United States opioid epidemic disproportionately harms individuals from disadvantaged backgrounds, with certain populations facing a significantly higher burden of opioid use disorder (OUD)-related harms¹. Black and Latinx individuals, as well as those involved in the justice system, experience disproportionately high rates of overdose and adverse health outcomes associated with opioid use²⁻⁴. These disparities were further exacerbated by the COVID-19 pandemic, during which opioid overdose mortality rates were markedly higher among Black, American Indian, and Alaska Native individuals compared to White individuals^{5,6}. In Missouri, drug overdose remains the leading cause of death among adults aged 18 to 44, with opioids involved in over 70% of these fatalities in 2023. In addition, Missouri's opioid overdose death rate at 23.03 per 100,000 in 2023, but Black males have experienced a significantly higher rate of 100.98 per 100,000—nearly four times higher⁷.

Despite the proven effectiveness of medications for opioid use disorder (MOUD) in reducing overdose risk, individuals from disadvantaged backgrounds remain less likely to receive these treatments⁸⁻¹⁰. Social determinants of health (SDoH), such as economic instability, healthcare access, and educational attainment, significantly shape disparities in OUD prevalence and treatment accessibility¹¹.

However, because individual-level SDoH are not well captured in electronic health records (EHRs), particularly in structured EHR data, it is challenging to assess and monitor their impact on care delivery and health outcomes¹²⁻¹⁴. Prior work has proposed linking EHR data with geospatial data sources¹⁵, such as census data, and using neighborhood-level SDoH as a proxy for evaluating the impact of SDoH on treatment allocation and health outcomes¹⁶. However, few studies have implemented this data linkage workflow and systematically assessed the impact of a broad panel of SDoH on OUD patients^{17,18}.

Our study acknowledges several informatics challenges identified by Gianfrancesco and Goldstein (2021) regarding EHR-based research¹⁹. This study aims to assess the impact of census-tract-level SDoH on the distribution of OUD and the allocation of MOUD among OUD patients in the St. Louis metropolitan area. By integrating EHRs with geospatial data, we seek to evaluate the geographic distribution of OUD cases in relation to socioeconomic disadvantage at the census-tract level, assess disparities in MOUD allocation based on demographic and

neighborhood-level SDoH factors, and identify spatial patterns of treatment inequities. This approach will provide novel insights into the impact of SDoH on OUD care and potential leading health outcomes, offering an evidence-based foundation for more targeted and timely interventions that promote equitable access to MOUD in high-risk communities.

Methods

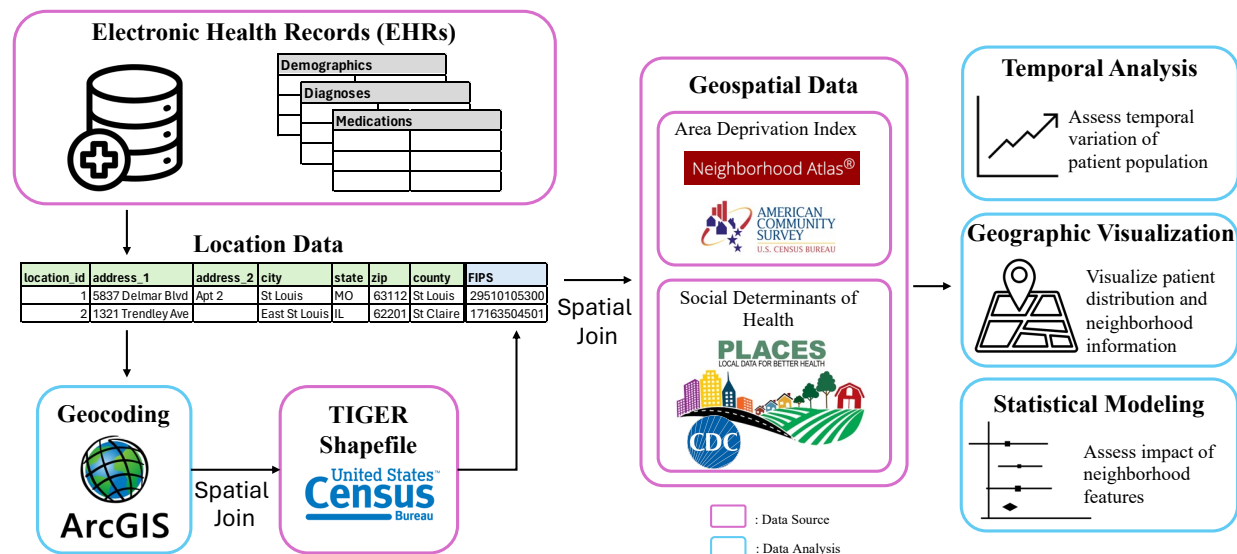


Figure 1. The workflow of integrating census data with electronic health records for health disparity research.

Data Sources. This study was approved by the Institutional Review Board of Washington University in St. Louis (Protocol #202406102). Informed consent was waived due to the retrospective nature of the study. The EHR data come from BJC HealthCare EHR database, including over 5.7 million patients across 15 hospitals which primarily located in the greater St. Louis, southern Illinois and mid-Missouri regions, serving the health care needs of urban, suburban, and rural communities. Data were transformed to the Observational Medical Outcomes Partnership (OMOP) Common Data Model (CDM) v5.3. We included patients with complete basic demographic information including sex, date of birth, residential location, and race/ethnicity data from January 1st, 2000, to December 31st, 2024.

The SDoH factors were extracted from Population Level Analysis and Community Estimates (PLACES) database. PLACES, developed by the Centers for Disease Control and Prevention (CDC) in collaboration with the Robert Wood Johnson Foundation (RWJF) and the CDC Foundation, provides estimates for over 40 health indicators, including *9 social determinants of health (SDoH)* derived from the 5-year American Community Survey (ACS). PLACES provides data at county, place, census tract, and ZIP Code Tabulation Area levels across the United States.

The ADI information was obtained from the Neighborhood Atlas, developed by University of Wisconsin School of Medicine and Public Health²⁰. The ADI represents a composite measure that incorporates 17 socioeconomic indicators in the domains of poverty, education, housing, and employment at the census block group level. This validated index has been extensively used in health disparity research and provides a complete assessment of neighborhood disadvantage.

Geocoding and Geospatial Data Integration. We developed an end-to-end workflow to integrate geospatial data with EHRs. The workflow begins with geocoding patient addresses from the EHR using an institutional HIPAA-compliant ArcGIS server to convert addresses to geographic coordinates (latitude and longitude)²¹. The quality of each matched address was accessed by a matching score (0-100) that indicates the precision of the geocoding. Since the majority of our patient population live in MO and IL, we excluded patients who resided outside of these two states. We then matched these coordinates to Census Block Groups using Federal Information Processing System (FIPS) codes from 2020 US Census Bureau shapefiles.²² This geospatial mapping from patient addresses to FIPS codes was developed by Jiang et al³⁶. Finally, we integrated external geospatial datasets (PLACES and Neighborhood Atlas) to incorporate ADI and SDoH. The data integration process was performed within the WashU-

licensed Azure Databricks cloud environment to best protect patient privacy. The enriched dataset includes ADI and SDoH at the census tract level, in addition to existing variables in the EHR.

Study Population. To capture all diagnoses of OUD and opioid overdose and corresponding MOUD prescription restricted to the same visit occurrence, we used the International Classification of Diseases, Ninth Revision, Clinical Modification (ICD-9-CM) International Statistical Classification of Diseases, Tenth Revision, Clinical Modification (ICD-10-CM) diagnosis code for OUD and opioid overdose and RxNorm code to capture MOUD receipts. Individuals were categorized as having an OUD diagnosis for a particular year if they obtained at least one diagnoses during outpatient, inpatient, or emergency room (ER) care at any BJC HealthCare facility.

All patient information was collected from January 1st, 2000, to December 31st, 2024. Demographic characteristics included sex (male, female), age group (<18, 18-34, 35-49, 50-64, and >64 years), and race (White, Black or African American, Other), where other race includes American Indian or Alaska Native, Asian, Native Hawaiian or Other Pacific Islander, and Unknown. Age was defined as patient's age at the time of first OUD or opioid overdose diagnosis in the EHR, since MOUD receipts were restricted to the same visit occurrence age was also calculated with the condition start time. Comorbidities were track through the study period.

Statistical Analysis. We first examined the annual prevalence of OUD and opioid overdose diagnoses from 2000 to 2024, stratified by sex, age group, and race to assess disparities. Second, we created geospatial maps to visualize the geospatial distribution of OUD and opioid overdose cases in both the MO and color-coded the census tracts by ADI. We have generated geospatial maps at different scale (state-level vs city-level) (Figure 3). To examine the associations between patient characteristics, census-tract-level SDoH, ADI, and type or receipt of MOUD, we conducted logistic regression analyses. Crude odds ratios (ORs) were estimated using univariable models, and adjusted ORs were obtained from multivariable logistic regression models including all covariates. The ORs and 95% confidence intervals (CIs) were reported in Table 2. For SDoH variables, ORs reflect the change in odds of receiving MOUD per 10 percentage point increase in the corresponding census-tract-level indicator. Statistical significance was denoted using asterisks (* $p < 0.05$, ** $p < 0.001$). Statistical analyses were conducted in R.

Results

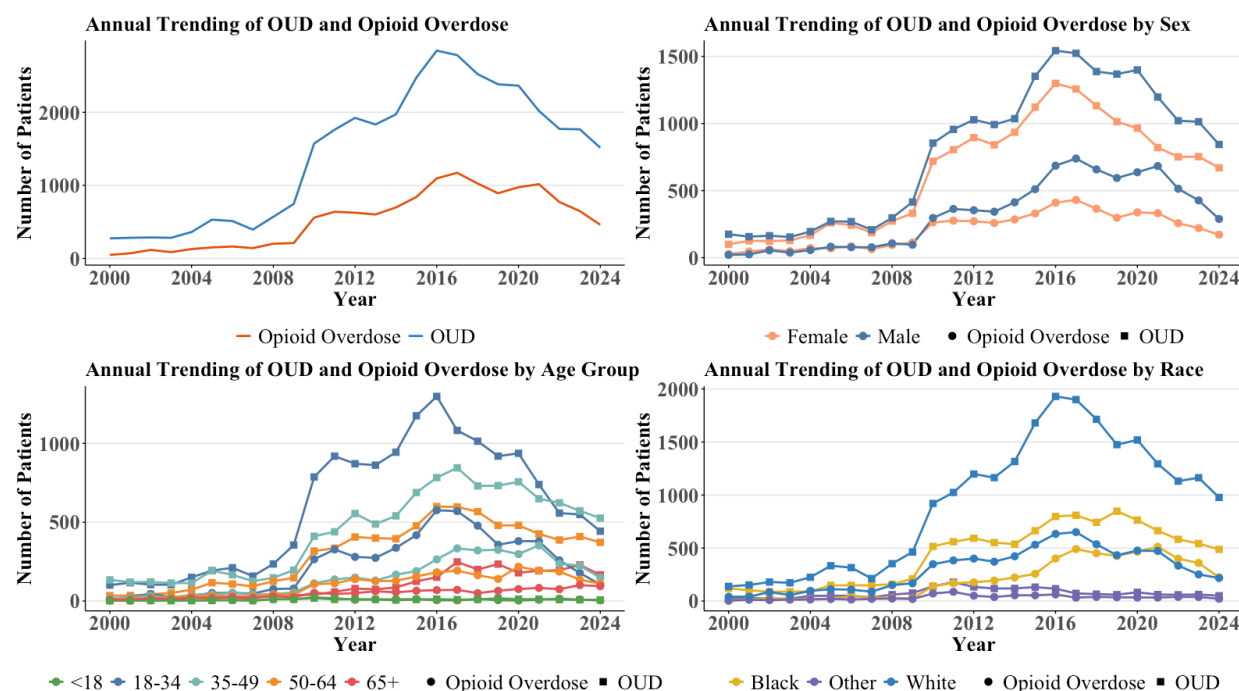


Figure 2. Temporal trends in opioid use disorder (OUD) and opioid overdoses among patients in BJC HealthCare from 2000 to 2024, stratified by sex, age group, race, and area deprivation index (ADI).

Our study identified 35,742 patients who have had at least one diagnosis of OUD, 13,345 patients who have had at least one diagnosis of opioid overdose, and 10,755 patients who have had at least one prescription of MOUD between 01/01/2000 and 12/31/2024 based on EHRs from BJC HealthCare.

Temporal Trends of OUD and Opioid Overdose. As shown in Figure 1, the number of patients diagnosed with OUD and opioid overdose steadily increased from 2000, peaking around 2016-2017 before declining in recent years. Specifically, OUD cases rose sharply after 2010, reaching a peak of 2,843 in 2016, followed by a gradual decline through 2024 to 1,225, representing about a 56.9% decline. A similar pattern was observed for opioid overdose, with a peak of 1,131 cases in 2017 and a noticeable decrease thereafter to 399 in 2024.

Sex-stratified trends indicate that males consistently exhibited higher rates of both OUD and overdose compared to females, although the gap narrowed slightly over time. Age-stratified trends revealed that individuals aged 18–34 consistently had the highest number of OUD and overdose cases throughout the study period, followed by those aged 35–49. All age groups showed similar temporal patterns with peaks around 2017, aligning with the overall trend. Race-stratified trends showed White patients accounted for the majority of OUD and overdose cases in the local healthcare system, with a sharp increase after 2010 and a peak around 2017.

Geographical Distribution of OUD and Opioid Overdose. We examined the geographic distribution of OUD and opioid overdose cases within the BJC Healthcare across the St. Louis metropolitan area using geospatial mapping (Figure 3). The maps illustrate a striking concentration of cases in census tracts with high ADI scores (7–10), which represent the most socioeconomically disadvantaged communities. These high-ADI areas are primarily concentrated within the central, northern and northeastern sectors of the St. Louis metropolitan area. In contrast, low-ADI areas, largely found in the western sectors, show markedly fewer cases. By 2024, high-ADI regions accounted for 58.42% of all OUD cases ($n = 20,543$) and 57.31% of overdose cases ($n = 7,648$). In contrast, low-ADI regions represented only 18.66% of OUD cases ($n = 6,561$) and 19.87% of overdose cases ($n = 2,651$).

The geographic overlap between OUD and overdose cases is substantial, especially in the northern sector of the metropolitan area, underscoring a critical area of public health concern. This overlap suggests that specific neighborhoods—many of which fall within the most deprived ADI range—bear a disproportionate burden of opioid-related harms and warrant targeted intervention.

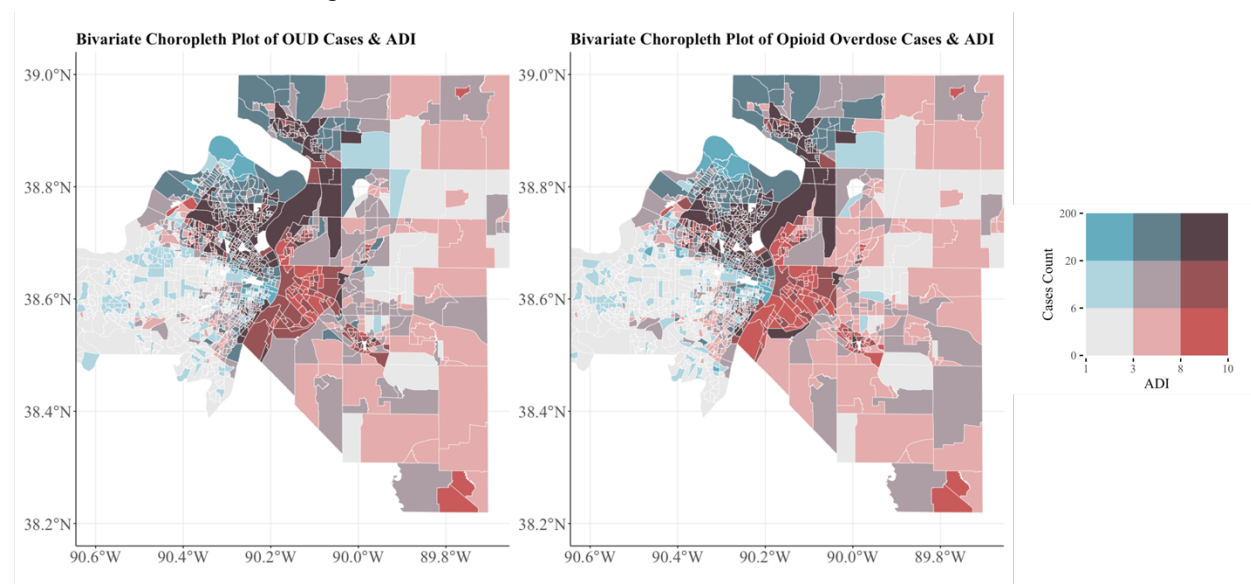


Figure 3. Geographic distribution of opioid use disorder and opioid overdose patients by census tract in the St. Louis metropolitan area. The bivariate choropleth map displays the intersection of area deprivation index (ADI, red channel) and the number of patients diagnosed with opioid use disorder or opioid overdose (blue channel). Darker shades indicate higher values—greater deprivation (red) and/or more patients (blue). Purple hues reflect areas with both high deprivation and high patient counts.

Disparities in the Receipt of MOUD. Table 1 summarizes the demographic, clinical, and social characteristics of patients with OUD stratified by type of MOUD received. Among the 10,755 patients who received MOUD, buprenorphine was the most commonly prescribed treatment (55.8%), followed by methadone (39.8%), with fewer patients receiving naltrexone (2.2%) or multiple MOUDs (2.2%). Most MOUD recipients were White (66.5%), aged 18–49 (79.0%), and male (52.2%).

Two multivariable regression models were fitted to explore disparities in treatment. The first model assessed predictors of receiving **methadone vs. buprenorphine**. After accounting for covariates, several significant associations were identified (Table 2). Individuals aged 35-49 were less likely to be administered methadone compared to buprenorphine (adjusted OR: 0.78, 95% CI: 0.71-0.86), as were male patients (adjusted OR: 0.71, 95% CI: 0.65-0.77). Several SDoH were found to significantly influence treatment patterns. Patients living in areas with higher housing crowding (adjusted OR: 0.95, 95% CI: 0.91-0.99), higher proportions of adults without high school diplomas (adjusted OR: 0.90, 95% CI: 0.85-0.95), and greater poverty (adjusted OR: 0.96, 95% CI: 0.91-1.00) were less likely to receive methadone. Conversely, patients from areas with higher proportions of racial or ethnic minorities (adjusted OR: 1.19, 95% CI: 1.10-1.28) and single-parent households (adjusted OR: 1.06, 95% CI: 1.00-1.12) were more likely to receive methadone. Patients with comorbid chronic pain (adjusted OR: 1.21, 95% CI: 1.11-1.32) and substance abuse (adjusted OR: 1.19, 95% CI: 1.06-1.33) were also significantly more likely to be treated with methadone.

Table 1. Characterization of patients with opioid use disorder, stratified by the type of medication for opioid use disorder (MOUD). Data are presented as counts and percentages for categorical variables and as means with 95% confidence intervals for continuous social determinant indicators.

	Buprenorphine (N = 5,996)	Methadone (N = 4,286)	Naltrexone (N = 236)	Multiple MOUD (N = 237)	No MOUD Received (N = 31,007)
Age					
< 18	7 (0.1%)	16 (0.4%)	2 (0.8%)	0 (0%)	318 (1.0%)
18 - 34	2,662 (44.4%)	2,007 (46.8%)	78 (33.1%)	87 (36.7%)	11,617 (37.5%)
35 - 49	2,178 (36.3%)	1,293 (30.2%)	92 (39.0%)	95 (40.1%)	8,934 (28.8%)
50 - 64	940 (15.7%)	780 (18.2%)	58 (24.6%)	47 (19.8%)	7,082 (22.8%)
65+	209 (3.5%)	190 (4.4%)	6 (2.5%)	8 (3.4%)	3,003 (9.7%)
Biological Sex					
Female	2,689 (44.8%)	2,263 (52.8%)	88 (37.3%)	108 (45.6%)	13,328 (43.0%)
Male	3,307 (55.2%)	2,023 (47.2%)	148 (62.7%)	129 (54.4%)	17,671 (57.0%)
Race					
White	4,047 (67.5%)	2,784 (65.0%)	155 (65.7%)	167 (70.5%)	19,324 (62.3%)
Black	1,784 (29.8%)	1,335 (31.1%)	72 (30.5%)	61 (25.7%)	9,718 (31.3%)
Other	165 (2.8%)	167 (3.9%)	9 (3.8%)	9 (3.8%)	1,966 (6.3%)
Area Deprivation Index					
Low (1-3)	907 (15.1%)	666 (15.5%)	52 (22.0%)	45 (19.0%)	6,229 (20.1%)
Medium (4-6)	1,367 (22.8%)	886 (20.7%)	61 (25.8%)	52 (21.9%)	7,039 (22.7%)
High (7-10)	3,630 (60.5%)	2,670 (62.3%)	121 (51.3%)	135 (57.0%)	17,237 (55.6%)
Social Determinants of Health (Mean, 95% CI)					
Housing crowding	1.8 (0.0-19.3)	1.6 (0.0-19.3)	1.8 (0.0-19.3)	1.8 (0.0-19.3)	1.7 (0.0-19.3)
Housing cost burden	29.4 (3.6-69.6)	29.6 (3.9-69.6)	28.9 (6.3-67.3)	29.8 (9.0-65.3)	28.9 (3.6-69.6)
No internet subscription	18.3 (0.4-59.6)	17.8 (0.4-59.6)	17.2 (0.4-52.2)	18.5 (1.0-52.7)	17.1 (0.0-59.6)
No high school diploma	11.6 (0.0-42.6)	11.0 (0.0-49.3)	10.8 (0.0-39.4)	10.9 (0.4-35.0)	10.9 (0.0-49.3)
Persons aged 65 years or older	16.3 (0.4-48.3)	16.1 (0.4-48.6)	15.5 (3.0-31.8)	16.3 (3.0-35.2)	16.3 (0.4-61.8)
Persons living below 150% of the poverty level	28.6 (0.2-80.8)	27.8 (0.2-80.8)	27.8 (0.2-80.8)	28.7 (2.1-74.2)	26.8 (0.2-84.7)
Persons of racial or ethnic minority status	38.4 (0.0-100.0)	41.2 (0.0-100.0)	39.2 (0.3-100.0)	40.3 (0.9-100.0)	39.0 (0.0-100.0)
Single-parent households	8.8 (0.0-33.6)	9.1 (0.0-33.6)	8.8 (0.0-33.6)	8.4 (0.0-33.6)	8.7 (0.0-42.8)

Unemployment	6.4 (0.0-30.5)	6.5 (0.0-30.5)	5.8 (0.0-30.5)	6.3 (0.0-25.8)	6.1 (0.0-36.6)
Comorbidities					
Depressive disorder	3,173 (52.9%)	2,349 (54.8%)	188 (79.7%)	126 (53.2%)	15,761 (50.8%)
Chronic pain	2,279 (38.0%)	1,827 (42.6%)	101 (42.8%)	115 (48.5%)	13,369 (43.1%)
Substance abuse	4923 (82.1%)	3,644 (85.0%)	209 (88.6%)	199 (84.0%)	24,029 (77.5%)

Table 2. Associations between patient characteristics, census-tract-level SDoH and ADI, and MOUD treatment type or receipt. Crude and adjusted odds ratios (ORs) were derived from univariable and multivariable logistic regression models, respectively. The 95% confidence intervals are shown in parenthesis. For SDoH variables, ORs represent the change in odds per 10 percentage point increase in the census-tract-level variable. Asterisks denote statistical significance (*p <0.05, **p<0.001).

	Methadone vs Buprenorphine		MOUD vs No MOUD	
	Crude ORs	Adjusted ORs	Crude ORs	Adjusted ORs
Age				
< 18	3.03 (1.29-7.90) *	3.25 (1.37-8.52) *	0.19 (0.12-0.28) **	0.20 (0.13-0.30) **
18 - 34	Reference	Reference	Reference	Reference
35 - 49	0.79 (0.72-0.86) **	0.78 (0.71-0.86) **	0.99 (0.94-1.04)	0.98 (0.93-1.04)
50 - 64	1.10 (0.98-1.23)	1.08 (0.96-1.21)	0.62 (0.59-0.66) **	0.62 (0.58-0.66) **
65+	1.21 (0.98-1.48)	1.21 (0.98-1.49)	0.33 (0.30-0.37) **	0.35 (0.31-0.39) **
Biological Sex				
Female	Reference	Reference	Reference	Reference
Male	0.73 (0.67-0.79) *	0.71 (0.65-0.77) **	0.82 (0.79-0.86) **	0.82 (0.79-0.86) **
Race				
White	Reference	Reference	Reference	Reference
Black or African American	1.09 (1.00-1.19)	0.92 (0.81-1.04)	0.90 (0.86-0.95) **	0.81 (0.76-0.87) **
Other	1.47 (1.18-1.83) **	1.45 (1.16-1.82) **	0.48 (0.43-0.54) **	0.47 (0.42-0.53) **
Area Deprivation Index				
Low (1-3)	Reference	Reference	Reference	Reference
Medium (4-6)	0.98 (0.83-1.16)	0.97 (0.81-1.17)	1.02 (0.93-1.12)	1.05 (0.96-1.16)
High (7-10)	0.92 (0.79-1.08)	1.03 (0.88-1.21)	1.20 (1.10-1.30) **	1.10 (1.01-1.20) *
Social Determinants of Health				
Housing crowding	0.77 (0.65-0.90) **	0.95 (0.91-0.99) *	1.07 (0.97-1.16)	0.98 (0.96-1.00)
Housing cost burden	1.02 (0.98-1.05)	1.05 (0.98-1.12)	1.04 (1.02-1.06) **	0.95 (0.92-0.99) *
No internet subscription	0.96 (0.92-1.00) *	1.04 (0.98-1.10)	1.09 (1.07-1.11) **	1.08 (1.04-1.12) **
No high school diploma	0.88 (0.83-0.93) **	0.90 (0.85-0.95) **	1.09 (1.05-1.12) **	0.95 (0.92-0.98) **
Persons aged 65 years or older	0.94 (0.88-1.01)	0.96 (0.91-1.00)	0.97 (0.94-1.01)	1.00 (0.98-1.03)
Persons living below 150% of the poverty level	0.97 (0.94-0.99) *	0.96 (0.91-1.00) **	1.07 (1.05-1.08) **	1.13 (1.08-1.19) **
Persons of racial or ethnic minority status	1.02 (1.01-1.03) **	1.19 (1.10-1.28) **	1.00 (1.00-1.01)	1.02 (0.97-1.06)
Single-parent households	1.08 (1.02-1.15) **	1.06 (1.00-1.12) *	1.06 (1.03-1.10) **	1.00 (0.97-1.03)
Unemployment	1.02 (0.95-1.11)	1.03 (0.98-1.08)	1.12 (1.07-1.17) **	1.02 (0.99-1.05)
Comorbidities				
Depressive disorder	1.08 (1.00-1.17)	0.94 (0.86-1.02)	1.15 (1.10-1.20) **	1.15 (1.09-1.21) **

Chronic pain	1.22 (1.12-1.32) **	1.21 (1.11-1.32) **	0.88 (0.85-0.92) **	0.92 (0.87-0.96) **
Substance abuse	1.24 (1.11-1.38) **	1.19 (1.06-1.33) **	1.47 (1.39-1.55) **	1.29 (1.22-1.37) **

The second model compared patients who **received any MOUD versus no treatment**. Older patients (ages 50+) were notably less likely to undergo treatment, as were males (adjusted OR: 0.82, 95% CI: 0.79-0.86). Racial disparities were observed, with Black/African American patients (adjusted OR: 0.81, 95% CI: 0.76-0.87) and patients classified under Other race (adjusted OR: 0.47, 95% CI: 0.42-0.53) less likely to receive treatments compared to White patients. Socioeconomic context played a notable role: individuals residing in areas with high ADI scores (7-10) were more likely to receive MOUD (adjusted OR: 1.10, 95% CI: 1.01-1.20). Regarding the impact of clinical factors, patients with depressive disorders (adjusted OR: 1.15, 95% CI: 1.09-1.21) and substance abuse issues (adjusted OR: 1.29, 95% CI: 1.22-1.37) were more likely to receive MOUD, whereas those with chronic pain were less likely (adjusted OR: 0.92, 95% CI: 0.87-0.96).

Discussion

A growing body of research highlights the potential of integrating social and environmental determinants of health from geospatial data and clinical data from EHRs to support real-world evidence of health disparity and equity research²³⁻²⁵. While EHRs excel in capturing longitudinal treatment and health outcomes, they often lack comprehensive information on SDoH. Conversely, geospatial data sources, such as U.S. Census Bureau surveys, provide rich SDoH insights but lack individual-level clinical detail. Our data integration workflow bridges this gap by integrating neighborhood-level SDoH from US census data into EHRs, which allow us to construct a multifaceted dataset to systematically investigate disparities in treatment allocation across diverse populations.

Our study revealed significant geographic disparities in both OUD and opioid overdose patient encounter counts and MOUD administering across the St. Louis metropolitan region, with a substantial concentration of OUD and overdose cases in highly disadvantaged areas over the 24-year study period. By integrating SDoH data with EHRs, we identified persistent disparity patterns that align with previous research showing how socioeconomic factors such as poverty and unemployment correlate with increased risk of opioid overdose²⁶⁻²⁹. Some of our findings, such as lower access to MOUD among black patients compared to white patients, are consistent with existing literature³⁰. Others, including unexpected sex differences in MOUD receipt and unique age-related access patterns, diverge from previous studies^{31,32}. For instance, a study conducted on the New York City population indicated that regions with elevated income levels exhibited higher rates of buprenorphine treatment, our analysis conducted in St. Louis demonstrated that a 10% increase in the population living below 150% of the poverty threshold corresponded with an increased likelihood of receiving buprenorphine³³. These inconsistent findings may reflect local provider practices, stigma, transportation barriers, or differential insurance coverage, underscoring the need to explore drivers of health disparity at both nation-level and local healthcare system-level.

Our workflow illustrates how integrating geospatial data with EHRs can uncover the impact of neighborhood-level social determinants of health (SDoH) on treatment disparities that may be overlooked by traditional EHR-based analyses. This geospatially enriched framework is readily adaptable to other healthcare systems and has broad potential for enhancing clinical decision-making, informing resource allocation, and guiding community outreach efforts.

First, this framework can be extended into a clinical decision support tool to support clinical decision-making at the point of care by embedding neighborhood-level SDoH indicators into the EHR interface. For example, clinicians could be alerted when a patient resides in a census tract with high deprivation, housing instability, or limited access to transportation. These contextual risk flags could inform personalized care plans, such as recommending longer-acting formulations of medications for opioid use disorder (MOUD), scheduling more frequent follow-ups, or involving case managers or social workers earlier in care. A related example is the Envirome Web Service (EWS), a clinical decision support tool developed for a large pediatric health system in the Kansas City region, which securely geocodes patient addresses in real time and links them to publicly available neighborhood-level data³⁴. Deployment of the EWS at the point of care enabled providers to recognize contextual barriers—such as food deserts or transportation access—and incorporate this information into shared decision-making with families. Early findings from the EWS suggest that presenting environmental risk indicators in a simple, actionable format helps providers tailor their communication and connect patients with appropriate resources, ultimately supporting both improved patient experience and health outcomes. By making structural vulnerability visible in routine care settings, our framework can empower providers to address non-clinical barriers that might otherwise compromise treatment adherence and long-term recovery.

Second, this framework can inform resource allocation and program planning within the healthcare system. BJC HealthCare can use the insights from this analysis to identify treatment deserts and geographic hotspots where OUD prevalence is high but MOUD access is low. These findings can guide the strategic deployment of mobile clinics, satellite treatment sites, or telehealth programs to underserved areas. The framework also provides a data-driven basis for evaluating existing programs and prioritizing investments in new initiatives that target the most affected neighborhoods. Moreover, integrating these insights into health system planning tools could ensure that staffing, outreach, and service availability are aligned with geographic patterns of need.

Third, this framework can be leveraged to guide community outreach initiatives beyond the scope of the healthcare system. While healthcare system may have limited influence on addressing societal barriers, the identification of high-risk communities enables collaboration with local governments, nonprofits, and public health agencies. These partners could use hotspot data and neighborhood-level SDoH indicators to expand harm reduction services, launch public awareness campaigns, or provide transportation support programs³⁵. The collaboration between healthcare systems and community partners can foster a shared responsibility across sectors to address the structural drivers of the opioid epidemic.

Limitations. Our investigation was a retrospective cohort analysis using EHR data from a single local healthcare system, with clinical sites concentrated in Missouri and Illinois. Given EHR data often exhibit significant variations in geographic representativeness across different regions²³, the results should be interpreted as regional-level estimates and may not generalize to other parts of the country. However, our workflow was developed against OMOP, a widely adopted common data model by healthcare systems and by the federated research data network, observational health data sciences and informatics (OHDSI), we could apply the workflow for broader application across distributed clinical data networks such as OHDSI. A federated approach to replicate this analysis across multiple health systems would support disparities studies across multiple regions.

In addition, the inclusion of neighborhood-level SDoH in our study is still not exhaustive. Currently, SDoH and comorbidities related to MOUD accessibility suggest a potential link between social needs. However, the PLACES database lacks data on health-related social needs (HRSN) for the eastern region of the St. Louis metropolitan area, located in IL.

Furthermore, although neighborhood-level SDoH offer valuable contextual insight, they may not accurately reflect individual-level experiences, raising the potential for ecological fallacy. We could not determine from this analysis how much of the treatment gap is due to individual-level barriers, such as transportation limitations due to car possession. This limitation can be mitigated in future studies by incorporating individual-level SDoH from unstructured EHR data, such as clinical notes and questionnaires.

Lastly, the OMOP CDM v5.3 retains only patients' most recent address and does not capture address history. This limitation introduces potential bias due to patient mobility, particularly in longitudinal analyses. To address this issue, the OMOP Common Data Model could evolve to better support longitudinal geospatial analyses by incorporating address history with time stamps, which could address a critical gap in current common data model. With availabilities of longitudinal addresses would open new opportunities to study the impact of evolving treatment practices or policy changes on OUD prevalence and MOUD administering.

Conclusion

This study demonstrates the value of integrating geospatial social determinants of health data with electronic health records to identify and quantify disparities in opioid use disorder patient distribution and medication for treatment access. By applying a data integration and analytics workflow across EHR databases from a local healthcare systems. We unveiled the impact of neighborhood-level social determinants of health, race, sex, and age on disease distribution and medication allocation. Our findings highlight that even within a single metropolitan area, contextual social disadvantage remains strongly associated with both disease burden and treatment access. The analytic and informatics infrastructure developed in this study offers a generalizable model for future research on health disparities.

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References

1. 2022 national survey on drug use and health (NSDUH) releases | CBHSQ data [Internet]. [cited 2025 Mar 23]. Available from: <https://www.samhsa.gov/data/data-we-collect/nsduh-national-survey-drug-use-and-health/national-releases/2022>
2. Knighton JS, Stevens-Watkins D, Staton M, Pangburn K. Trends and mental health correlates of nonmedical opioid use among criminal justice-involved african American men. *Addictive Behaviors*. 2018 Oct 1;85:14–20.
3. Substance Abuse and Mental Health Services Administration. The Opioid Crisis and the Black/African American Population: An Urgent Issue. Publication No. PEP20-05-02-001. Rockville (MD): Office of Behavioral Health Equity, Substance Abuse and Mental Health Services Administration; 2020.
4. Gondré-Lewis MC, Abijo T, Gondré-Lewis TA. The opioid epidemic: A crisis disproportionately impacting black Americans and urban communities. *J Racial Ethn Health Disparities*. 2022 Sep 6;1–15.
5. Zhao Y, Liu Y, Lv F, He X, Ng WH, Qiu S, et al. Temporal trend of drug overdose-related deaths and excess deaths during the COVID-19 pandemic: A population-based study in the united states from 2012 to 2022. *eClinicalMedicine* [Internet]. 2024 Aug 1 [cited 2025 Mar 24];74. Available from: [https://www.thelancet.com/journals/eclinm/article/PIIS2589-5370\(24\)00331-6/fulltext](https://www.thelancet.com/journals/eclinm/article/PIIS2589-5370(24)00331-6/fulltext)
6. Jones AA, Santos-Lozada AR, Perez-Brumer A, Latkin C, Shoptaw S, El-Bassel N. Age-specific disparities in fatal drug overdoses highest among older black adults and American Indian/alaska native individuals of all ages in the united states, 2015–2020. *Int J Drug Policy*. 2023 Apr;114:103977.
7. Drug overdose dashboard - fatal overdoses | missouri department of health & senior services [Internet]. [cited 2025 Jan 16]. Available from: <https://health.mo.gov/data/opioids/>
8. Barnett ML, Meara E, Lewinson T, Hardy B, Chyn D, Onsando M, et al. Racial inequality in receipt of medications for opioid use disorder. *N Engl J Med*. 2023 May 11;388(19):1779–89.
9. Goedel WC, Shapiro A, Cerdá M, Tsai JW, Hadland SE, Marshall BDL. Association of racial/ethnic segregation with treatment capacity for opioid use disorder in counties in the united states. *JAMA Netw Open*. 2020 Apr 22;3(4):e203711.
10. Gibbons JB, Harris SJ, Solomon KT, Sugarman O, Hardy C, Saloner B. Increasing overdose deaths among black Americans: A review of the literature. *The Lancet Psychiatry*. 2023 Sep 1;10(9):719–26.
11. Joudrey PJ, Kolak M, Lin Q, Paykin S, Anguiano V, Wang EA. Assessment of community-level vulnerability and access to medications for opioid use disorder. *JAMA Netw Open*. 2022 Apr 19;5(4):e227028.
12. Blanco C, Wiley TRA, Lloyd JJ, Lopez MF, Volkow ND. America’s opioid crisis: The need for an integrated public health approach. *Transl Psychiatry*. 2020 May 28;10(1):167.
13. Ciccarone D. The rise of illicit fentanyl, stimulants and the fourth wave of the opioid overdose crisis. *Curr Opin Psychiatry*. 2021 Jul 1;34(4):344–50.
14. McMaster R, Masarweh-Zawahri L, Flynn KC, Deo VS, Flannery DJ. Drug Overdose Death among Residents of Urban Census Tracts: How Granular Geographical Analyses Uncover Socioenvironmental Correlates in Cuyahoga County, Ohio. *J Urban Health* [Internet]. 2024 Dec 3 [cited 2025 Feb 14]; Available from: <https://link.springer.com/10.1007/s11524-024-00939-8>
15. Xie S, Himes BE. Approaches to link geospatially varying social, economic, and environmental factors with electronic health record data to better understand asthma exacerbations. *AMIA Annu Symp Proc*. 2018 Dec 5;2018:1561–70.
16. Rodriguez ER, Tonn T, Jafry M, Ahmed S, Cuglievan B, Livingston JA, et al. Neighborhood-level social determinants of health burden among adolescent and young adult cancer patients and impact on overall survival. *JNCI Cancer Spectr*. 2024 Jul 25;8(4):pkae062.
17. Amiri S, McDonell MG, Denney JT, Buchwald D, Amram O. Disparities in access to opioid treatment programs and office-based buprenorphine treatment across the rural-urban and area deprivation continua: A US nationwide small area analysis. *Value in Health*. 2021 Feb;24(2):188–95.
18. Lindner SR, Hart K, Manibusan B, McCarty D, McConnell KJ. State- and county-level geographic variation in opioid use disorder, medication treatment, and opioid-related overdose among medicaid enrollees. *JAMA Health Forum*. 2023 Jun 23;4(6):e231574.
19. Gianfrancesco MA, Goldstein ND. A narrative review on the validity of electronic health record-based research in epidemiology. *BMC Med Res Methodol*. 2021 Dec;21(1):234.
20. Kind AJH, Buckingham WR. Making neighborhood-disadvantage metrics accessible — the neighborhood atlas. *N Engl J Med*. 2018 Jun 28;378(26):2456–8. AND University of Wisconsin School of Medicine Public Health. 2022

- Area Deprivation Index v.4.0.1 [Internet]. Madison (WI): University of Wisconsin; 2022 [cited 2024 Dec 18]. Available from: <https://www.neighborhoodatlas.medicine.wisc.edu/>
21. Tips for improving geocoding quality—ArcGIS pro | Documentation. Accessed March 18, 2025. <https://pro.arcgis.com/en/pro-app/2.8/help/data/geocoding/tips-for-improving-geocoding-quality.htm>
22. Bureau UC. TIGER/line shapefiles [Internet]. Census.gov. [cited 2025 Mar 9]. Available from: <https://www.census.gov/geographies/mapping-files/2020/geo/tiger-line-file.html>
23. Xie SJ, Kapos FP, Mooney SJ, Mooney S, Stephens KA, Chen C, et al. Geospatial divide in real-world EHR data: Analytical workflow to assess regional biases and potential impact on health equity. *AMIA Jt Summits Transl Sci Proc.* 2023 Jun 16;2023:572–81.
24. Xie S, Greenblatt R, Levy MZ, Himes BE. Enhancing electronic health record data with geospatial information. *AMIA Jt Summits Transl Sci Proc.* 2017 Jul 26;2017:123–32.
25. Laranjo L, Rodrigues D, Pereira AM, Ribeiro RT, Boavida JM. Use of electronic health records and geographic information systems in public health surveillance of type 2 diabetes: A feasibility study. *JMIR Public Health Surveill.* 2016 Mar 17;2(1):e12.
26. Pear VA, Ponicki WR, Gaidus A, Keyes KM, Martins SS, Fink DS, et al. Urban-rural variation in the socioeconomic determinants of opioid overdose. *Drug and Alcohol Dependence.* 2019 Feb;195:66–73.
27. Altekruse SF, Cosgrove CM, Altekruse WC, Jenkins RA, Blanco C. Socioeconomic risk factors for fatal opioid overdoses in the united states: Findings from the mortality disparities in American communities study (MDAC). *PLoS One.* 2020 Jan 17;15(1):e0227966.
28. Van Draanen J, Tsang C, Mitra S, Karamouzian M, Richardson L. Socioeconomic marginalization and opioid-related overdose: A systematic review. *Drug and Alcohol Dependence.* 2020 Sep;214:108127.
29. Chichester K, Drawve G, Sisson M, McCleskey B, Dye DW, Cropsey K. Examining the neighborhood-level socioeconomic characteristics associated with fatal overdose by type of drug involved and overdose setting. *Addictive Behaviors.* 2020 Dec 1;111:106555.
30. Stahler GJ, Mennis J, Baron DA. Racial/ethnic disparities in the use of medications for opioid use disorder (MOUD) and their effects on residential drug treatment outcomes in the US. *Drug and Alcohol Dependence.* 2021 Sep;226:108849.
31. Filiatreau LM, Szlyk HS, Kasson E, Dickson MF, Banks D, Winograd R, et al. Attitudes and subjective norms regarding medication for opioid use disorder among individuals in treatment for opioid use disorder in the greater st. Louis area. *Journal of Drug Issues.* 2024 Feb 1;00220426241229972.
32. Finlay AK, Harris AHS, Timko C, Yu M, Smelson D, Stimmel M, et al. Disparities in access to medications for opioid use disorder in the veterans health administration. *J Addict Med.* 2021 Apr 1;15(2):143–9.
33. Hansen HB, Siegel CE, Case BG, Bertollo DN, DiRocco D, Galanter M. Variation in use of buprenorphine and methadone treatment by racial, ethnic and income characteristics of residential social areas in new york city. *J Behav Health Serv Res.* 2013 Jul;40(3):10.1007/s11414-013-9341–3.
34. Kane NJ, Wang X, Gerkovich MM, Breitzkreutz M, Rivera B, Kunchithapatham H, et al. The envirome web service: Patient context at the point of care. *Journal of Biomedical Informatics.* 2021 Jul;119:103817.
35. Stopka TJ, Larochelle MR, Li X, Bernson D, Li W, Ackerson LK, et al. Opioid-related mortality: Dynamic temporal and spatial trends by drug type and demographic subpopulations, Massachusetts, 2005–2021. *Drug and Alcohol Dependence.* 2023 May;246:109836.
36. Jiang X, Beaton MA, Gillberg J, Williams A, Natarajan K. Feasibility of Linking Area Deprivation Index Data to the OMOP Common Data Model. *AMIA Annu Symp Proc.* 2023;2022:587-595. 2023 Apr.